

The Obstruction Spectrum: A Consolidated Theorem–Proof Package

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Abstract

This document collects, in one place and with self-contained proofs, the theorems asserted in the abstract of *The Obstruction Spectrum of Weyl Context Families and the 2-adic Tower* (“Paper B”) together with the local-valuation theorem of the companion note. Each proof step is explicitly tagged [COMPLETE] (complete self-contained proof), [SKETCH: CITED] (sketch relying on a named external result, cited), or [GAP] (an open sub-lemma). The acceptance target is that every theorem in the Paper B abstract have a proof here, not merely a pointer to a script; where a step is genuinely inherited (notably the general- d attainment of Raussendorf–Okay–Zurel–Feldmann) or genuinely open, we say so. A Status table closes the document. The single most load-bearing external input is the Weyl–Heisenberg product law (Lemma 2), which we derive; everything in Part I is then combinatorial modular arithmetic. The one substantive residual gap is the general even- d attainment beyond the residue class $d \equiv 2 \pmod{4}$, which we prove in full, and beyond the explicit certificates at $d = 8, 16$: for $d \equiv 0 \pmod{4}$ in general we rely on [1].

Contents

1	Conventions and the standing hypotheses	2
	Part I. The Obstruction Spectrum	3
2	The Obstruction Spectrum Theorem	3
3	The commutator–carry form	4
4	Detection equivalence	6
5	The exact value-bit formula	7
6	Attainment and the classification	8
7	Depth Rigidity	10
	Part II. The Local Valuation Theorem	11

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8	Setup and the parity lemma	11
9	The cyclic transgression	11
10	The theorem	12
11	Status	13

1 Conventions and the standing hypotheses

Fix a local dimension $d \geq 2$ and a number of qudits $m \geq 1$. Phase space is $V = \mathbb{Z}^{2m}$ with coordinates written per site $v = (a_1, b_1, \dots, a_m, b_m)$; we reduce mod d when needed and take representatives in $\{0, \dots, d-1\}$. Write

$$q(v) = \sum_{i=1}^m a_i b_i, \quad \beta(u, v) = \sum_{i=1}^m u_{b,i} v_{a,i}, \quad \langle u, v \rangle = \beta(u, v) - \beta(v, u) = \sum_{i=1}^m (u_{a,i} v_{b,i} - u_{b,i} v_{a,i}),$$

so q is a $\mathbb{Z}$ -valued quadratic form, β a $\mathbb{Z}$ -bilinear form, and $\langle \cdot, \cdot \rangle$ the (alternating, $\mathbb{Z}$ -bilinear) symplectic form. Let $\omega = e^{2\pi i/d}$ and $\tau = e^{i\pi/d}$, so $\tau^2 = \omega$, τ has multiplicative order $2d$, and $\tau^d = e^{i\pi} = -1$. On a single qudit $X|j\rangle = |j+1 \bmod d\rangle$, $Z|j\rangle = \omega^j|j\rangle$; the *symmetric (Weyl) lift* is

$$W(v) = \tau^{-q(v)} X^{a_1} Z^{b_1} \otimes \dots \otimes X^{a_m} Z^{b_m}.$$

Definition 1 (Contexts, families, certificates). A *context* $C = (v_1, \dots, v_n)$ is a finite word (repetitions allowed) of *pairwise commuting* vectors, $\langle v_i, v_j \rangle \equiv 0 \pmod{d}$, that is *closed*: $\sum_i v_i \equiv 0 \pmod{d}$. Its product is a scalar, $\prod_i W(v_i) = \omega^{s(C)} I$ for a well-defined $s(C) \in \mathbb{Z}_d$ (see Lemma 13; for now $s(C)$ names the ω -exponent when it exists). A *family* $\mathcal{F} = \{C_1, \dots, C_k\}$ has *incidence (multiplicity) matrix* $A \in \mathbb{Z}^{k \times N}$, $A_{r,v}$ = number of times observable v occurs in C_r , over the set of N distinct observables, and *phase vector* $s = (s(C_r))_r \in \mathbb{Z}_d^k$. A *certificate* is $\lambda \in \mathbb{Z}_d^k$ with $\lambda^\top A \equiv 0 \pmod{d}$; its *value* is $S = \lambda^\top s \bmod d$. A nonzero value certifies that no $\mathbb{Z}_d$ -valued noncontextual assignment is consistent with $\mathcal{F}$ (an all-versus-nothing obstruction).

Scope declared once. Throughout Part I the *only* hypotheses on a family are those in Definition 1: each context is closed and internally commuting. We *explicitly permit*: (i) arbitrary multiplicities and repeated observables inside a context (a letter may recur); (ii) nonmaximal contexts (a context need not be a maximal commuting set); (iii) disconnected incidence structures (the observable–context bipartite graph may have several components); (iv) certificates λ with arbitrary $\mathbb{Z}_d$ entries, not only 0/1. Where a result needs more (e.g. “pure fiber pool” in Theorem 18) it is stated locally.

Lemma 2 (Weyl product law). *For all $u, v \in \mathbb{Z}^{2m}$,*

$$W(u)W(v) = \tau^{c(u,v)} W((u+v) \bmod d), \quad c(u, v) = -q(u) - q(v) + 2\beta(u, v) + q((u+v) \bmod d),$$

with $c(u, v)$ read modulo $2d$. Consequently $W(u)W(v) = \omega^{\langle u, v \rangle} W(v)W(u)$, so u, v commute iff $\langle u, v \rangle \equiv 0 \pmod{d}$.

Proof. [COMPLETE] It suffices to treat one qudit and tensor. On one qudit $ZX = \omega XZ$, hence $Z^b X^{a'} = \omega^{a'b} X^{a'} Z^b$ and $X^a Z^b X^{a'} Z^{b'} = \omega^{a'b} X^{a+a'} Z^{b+b'}$. Because $X^d = Z^d = I$, $X^{a+a'} = X^{s_a}$,

$Z^{b+b'} = Z^{s_b}$ with $s = (u + v) \bmod d$; and $X^{s_a} Z^{s_b} = \tau^{q(s)} W(s)$. Collecting the τ -powers, using $\omega^{a'b} = \tau^{2a'b} = \tau^{2\beta(u,v)}$ (single qudit $\beta(u,v) = b a'$),

$$W(u)W(v) = \tau^{-q(u)-q(v)} \tau^{2\beta(u,v)} \tau^{q(s)} W(s),$$

which is the claim. Swapping u, v changes the exponent by $2(\beta(u,v) - \beta(v,u)) = 2\langle u, v \rangle$, and $\tau^{2\langle u, v \rangle} = \omega^{\langle u, v \rangle}$. The multi-qudit statement is the tensor product; all three forms $q, \beta, \langle \cdot, \cdot \rangle$ are the site-wise sums, so the identity adds over sites. $\square$

Lemma 3 (Telescoped context phase). *For a context $C = (v_1, \dots, v_n)$ let $P_i = (v_1 + \dots + v_{i-1}) \bmod d$ (so $P_1 = 0$). Then $\prod_{i=1}^n W(v_i) = \tau^T W(\sum_i v_i \bmod d)$ with*

$$T = -\sum_i q(v_i) + 2\sum_i \beta(P_i, v_i) \equiv -\sum_i q(v_i) + 2\sum_{j<i} \beta(v_j, v_i) \pmod{2d}.$$

If C is closed then the product is $\tau^T I$.

Proof. [COMPLETE] Induct with Lemma 2: multiplying the partial product $\tau^{\dots} W(P_i)$ by $W(v_i)$ adds $c(P_i, v_i) = -q(P_i) - q(v_i) + 2\beta(P_i, v_i) + q(P_{i+1})$, where $P_{i+1} = (P_i + v_i) \bmod d$. Summing $i = 1, \dots, n$, the pair $-q(P_i) + q(P_{i+1})$ telescopes to $q(P_{n+1}) - q(P_1) = q(0) - q(0) = 0$. This leaves $T = -\sum_i q(v_i) + 2\sum_i \beta(P_i, v_i)$. Finally $\beta(P_i, v_i) \equiv \beta(\sum_{j<i} v_j, v_i) = \sum_{j<i} \beta(v_j, v_i) \pmod{d}$ by $\mathbb{Z}$ -bilinearity (replacing P_i by the unreduced prefix changes β by a multiple of d), so $2\sum_i \beta(P_i, v_i) \equiv 2\sum_{j<i} \beta(v_j, v_i) \pmod{2d}$. $\square$

Part I. The Obstruction Spectrum

2 The Obstruction Spectrum Theorem

Theorem 4 (Obstruction Spectrum). *For every d , every m , every family $\mathcal{F}$ satisfying Definition 1, and every certificate λ , the value obeys*

$$2S \equiv 0 \pmod{d}.$$

Hence $S \in \{0, d/2\}$ for even d , and $S = 0$ for odd d . Thus the set $H(d, m)$ of achievable state-independent values satisfies $H \subseteq \{0, d/2\}$ (even d) and $H = \{0\}$ (odd d). The bound holds verbatim in the presence of multiplicities, repeated observables, nonmaximal contexts, and disconnected incidence.

Proof. [COMPLETE] Fix a closed commuting context C . Reduce the exact identity of Lemma 3 modulo d :

$$2s(C) \equiv -\sum_i q(v_i) + 2\sum_{j<i} \beta(v_j, v_i) \pmod{d}. \quad (1)$$

(Here $\tau^T = \omega^{s(C)}$ means $2s(C) \equiv T \pmod{2d}$; we use it mod d .) Now set $\sigma = \sum_i v_i$. Bilinearity gives

$$\beta(\sigma, \sigma) = \sum_i \beta(v_i, v_i) + \sum_{i<j} (\beta(v_i, v_j) + \beta(v_j, v_i)) = \sum_i q(v_i) + \sum_{i<j} (\beta(v_i, v_j) + \beta(v_j, v_i)),$$

using $\beta(v, v) = q(v)$. On commuting pairs $\beta(v_i, v_j) \equiv \beta(v_j, v_i) \pmod{d}$ (because their difference is $\langle v_i, v_j \rangle \equiv 0$), so $\beta(\sigma, \sigma) \equiv \sum_i q(v_i) + 2 \sum_{i < j} \beta(v_j, v_i) \pmod{d}$. Closedness gives $\sigma \equiv 0 \pmod{d}$, i.e. $\sigma = dt$ for an integer vector t , whence $\beta(\sigma, \sigma) = d^2 \beta(t, t) \equiv 0 \pmod{d}$. Therefore

$$2 \sum_{i < j} \beta(v_j, v_i) \equiv - \sum_i q(v_i) \pmod{d}. \quad (2)$$

Substituting (2) into (1),

$$2s(C) \equiv -2 \sum_{v \in C} q(v) \pmod{d}. \quad (3)$$

Equation (3) says $2s \equiv -2Aq \pmod{d}$ as vectors over contexts, where $q = (q(v))_v$. For any certificate $(\lambda^\top A \equiv 0)$,

$$2S = \lambda^\top(2s) \equiv -2(\lambda^\top A)q \equiv 0 \pmod{d}.$$

For even d , $2S \equiv 0$ forces $S \in \{0, d/2\}$; for odd d , 2 is invertible so $S = 0$. None of the steps used maximality, distinctness of observables, connectivity, or a bound on multiplicities; C was an arbitrary closed commuting word and λ an arbitrary left-kernel vector. $\square$

Corollary 5 (Odd d is trivial, sharply). *For odd d , dividing (3) by the invertible 2 gives $s(C) \equiv -\sum_{v \in C} q(v) \pmod{d}$ outright, so $s \equiv -Aq$ is an exact coboundary and every certificate value is 0.* [COMPLETE]

Corollary 6 (The obstruction is one bit). *For even d , (3) gives $s(C) + \sum_{v \in C} q(v) \equiv 0 \pmod{d/2}$, so*

$$b(C) := \frac{s(C) + \sum_{v \in C} q(v)}{d/2} \pmod{2} \in \mathbb{Z}_2$$

is well defined, and $s(C) \equiv -\sum_{v \in C} q(v) + \frac{d}{2} b(C) \pmod{d}$. For any certificate, $S \equiv \frac{d}{2}(\bar{\lambda} \cdot b) \pmod{d}$ with $\bar{\lambda} = \lambda \pmod{2}$; only $\lambda \pmod{2}$ carries value, and an attaining certificate must have odd $\bar{\lambda} \cdot b$. [COMPLETE]

Proof. [COMPLETE] $b(C)$ is well defined by (3). For a certificate, $S = \lambda^\top s \equiv -\lambda^\top Aq + \frac{d}{2} \lambda^\top b \equiv \frac{d}{2}(\lambda \cdot b) \pmod{d}$, and $\frac{d}{2}(\lambda \cdot b) \pmod{d}$ depends only on $\lambda \cdot b \pmod{2} = \bar{\lambda} \cdot b$. $\square$

Remark 7 (No obstruction of order > 2). Corollary 6 shows the value always lies in the order-2 subgroup $\{0, d/2\} \subset \mathbb{Z}_d$, whatever the multiplicity structure. If each observable lies in r contexts, the value group is still $\subseteq \{0, d/2\}$, independent of r : there are no genuine order- d “qudit parity proofs”. [COMPLETE]

3 The commutator–carry form

Theorem 8 (Carry form). *Exactly, $2s(C) \equiv -\sum_i q(v_i) + 2 \sum_{j < i} \beta(v_j, v_i) \pmod{2d}$ (Lemma 3), and for even d the obstruction bit of Corollary 6 equals the total commutator carry:*

$$b(C) \equiv \sum_{\{u, v\} \subseteq C} \frac{\langle u, v \rangle}{d} \pmod{2},$$

the sum over unordered pairs of positions in the word C . The entire even- d obstruction theory is thus combinatorial: no phases, no matrices, only symplectic carries $\langle u, v \rangle/d \in \mathbb{Z}$.

Proof. [COMPLETE] We derive the carry expression from the cocycle formula rather than assert it. Start from the polarization computed inside Theorem 4, but keep the symmetric/antisymmetric split exactly. Using $\beta(v_i, v_j) = \beta(v_j, v_i) + \langle v_i, v_j \rangle$,

$$\beta(\sigma, \sigma) = \sum_i q(v_i) + \sum_{i < j} (2\beta(v_j, v_i) + \langle v_i, v_j \rangle) = \left[\sum_i q(v_i) + 2 \sum_{i < j} \beta(v_j, v_i) \right] + \sum_{i < j} \langle v_i, v_j \rangle.$$

Hence, with $\sigma = dt$ so $\beta(\sigma, \sigma) = d^2\beta(t, t)$,

$$\sum_i q(v_i) + 2 \sum_{i < j} \beta(v_j, v_i) = d^2\beta(t, t) - \sum_{i < j} \langle v_i, v_j \rangle. \quad (4)$$

Now compute $b(C)$ from its definition. By Corollary 6, $\frac{d}{2}b(C) \equiv s(C) + \sum q(v_i) \pmod{d}$, so $db(C) \equiv 2s(C) + 2 \sum q(v_i) \pmod{2d}$, and by (1) lifted to the exact mod- $2d$ identity of Lemma 3,

$$db(C) \equiv \left(- \sum q(v_i) + 2 \sum_{j < i} \beta(v_j, v_i) \right) + 2 \sum q(v_i) = \sum q(v_i) + 2 \sum_{i < j} \beta(v_j, v_i) \pmod{2d}.$$

Insert (4): $db(C) \equiv d^2\beta(t, t) - \sum_{i < j} \langle v_i, v_j \rangle \pmod{2d}$. Each commuting pair has $\langle v_i, v_j \rangle \equiv 0 \pmod{d}$, so $\langle v_i, v_j \rangle/d \in \mathbb{Z}$ and we may divide by d :

$$b(C) \equiv d\beta(t, t) - \sum_{i < j} \frac{\langle v_i, v_j \rangle}{d} \pmod{2}.$$

For even d the term $d\beta(t, t) \equiv 0 \pmod{2}$, and $-1 \equiv 1 \pmod{2}$, giving $b(C) \equiv \sum_{i < j} \langle v_i, v_j \rangle/d \pmod{2}$. Diagonal “pairs” from a repeated letter contribute $\langle v, v \rangle/d = 0$, so the formula is insensitive to repetitions, as claimed. $\square$

Corollary 9 (Closed-form contextuality criterion; cycle = self-pairing). *Let d be even and let A (over $\mathbb{F}_2$) be the incidence matrix, $b = (b(C_r))_r$ the carry vector. Call a $\mathbb{Z}_2$ left-kernel vector λ ($\lambda^\top A \equiv 0 \pmod{2}$) a cycle. Then*

$$\lambda \cdot b \equiv Q(\lambda) := \sum_{a < b \text{ in the observable multiset of } \lambda} \frac{\langle v_a, v_b \rangle}{d} \pmod{2},$$

and the family carries a $\mathbb{Z}_2$ all-versus-nothing obstruction iff some cycle has $Q(\lambda)$ odd.

Proof. [COMPLETE] Write $Q(\lambda) = \text{intra} + \text{inter}$ over pairs inside a single selected context and pairs across two selected contexts. By definition $\text{intra} = \sum_{r: \lambda_r=1} \sum_{i < j \in C_r} \langle v_i, v_j \rangle/d = \lambda \cdot b$ (Theorem 8). For the inter term, bilinearity gives $\sum_{u \in C_a, v \in C_b} \langle u, v \rangle = \langle S_a, S_b \rangle$ with $S_r = \sum_{v \in C_r} v = d t_r$ (closedness), so $\langle S_a, S_b \rangle = d^2 \langle t_a, t_b \rangle$ and $\text{inter} = \sum_{a < b} d \langle t_a, t_b \rangle \equiv 0 \pmod{2}$ for even d . Hence $Q(\lambda) = \lambda \cdot b$. The obstruction is $b \notin \text{rowspan}_{\mathbb{F}_2}(A)$, i.e. some left-kernel λ pairs oddly with b . $\square$

Proposition 10 (The exact $\mathbb{Z}_d$ criterion; Q is its mod-2 shadow). *Let M be the $\mathbb{Z}_d$ incidence matrix and $\gamma \in \mathbb{Z}_d^k$ the context-scalar vector. By the Fredholm alternative over $\mathbb{Z}_d$ (Lemma 11), $\mathcal{F}$ is state-independent contextual iff some $\mathbb{Z}_d$ -cycle λ ($\lambda^\top M \equiv 0 \pmod{d}$) has $\lambda \cdot \gamma \not\equiv 0 \pmod{d}$. The odd- Q test of Corollary 9 is the mod-2 reduction: a $\mathbb{Z}_d$ -cycle reduces to a $\mathbb{Z}_2$ -cycle (so odd- Q is sound), but a $\mathbb{Z}_2$ -cycle need not lift, so odd- Q can be incomplete on engineered minimal supports. [COMPLETE] for soundness; the empirical faithfulness on Weyl families (that the $\mathbb{Z}_2$ shadow and the $\mathbb{Z}_d$ obstruction agree except on engineered supports) is a verified phenomenon, not proved here [GAP].*

4 Detection equivalence

We first isolate the pure linear algebra, then feed in the Spectrum Theorem.

Lemma 11 (Fredholm alternative / double annihilator over $\mathbb{Z}_d$). *Let $A \in \mathbb{Z}^{k \times N}$ and $s \in \mathbb{Z}^k$. Exactly one of the following holds:*

- (a) *there is $\gamma \in \mathbb{Z}_d^N$ with $A\gamma \equiv s \pmod{d}$ (a noncontextual assignment);*
- (b) *there is $\lambda \in \mathbb{Z}_d^k$ with $\lambda^\top A \equiv 0 \pmod{d}$ and $\lambda^\top s \not\equiv 0 \pmod{d}$ (a certificate).*

Moreover in case (b) an explicit λ is produced constructively from the Smith Normal Form, valid for composite d .

Proof. [COMPLETE] *Mutual exclusion.* If both held, $\lambda^\top s \equiv \lambda^\top A\gamma \equiv 0$, contradiction. *Exhaustiveness with construction.* Compute the Smith Normal Form over $\mathbb{Z}$: $UAV = D = \text{diag}(g_1, \dots, g_r, 0, \dots, 0)$ with $U \in \text{GL}_k(\mathbb{Z})$, $V \in \text{GL}_N(\mathbb{Z})$, $g_1 \mid \dots \mid g_r$. Put $s' = Us$. Since U, V are unimodular they are invertible mod d , so $A\gamma \equiv s \pmod{d}$ is solvable iff $D\gamma' \equiv s' \pmod{d}$ is (with $\gamma' = V^{-1}\gamma$), which decouples into $g_i\gamma'_i \equiv s'_i \pmod{d}$ ($i \leq r$) and $0 \equiv s'_i \pmod{d}$ ($i > r$). The scalar congruence $g_i\gamma'_i \equiv s'_i \pmod{d}$ is solvable iff $\gcd(g_i, d) \mid s'_i$; the $i > r$ rows are solvable iff $d \mid s'_i$ (the case $\gcd(0, d) = d$).

If (a) fails, some index i violates its condition. Set $\mu^\top = \frac{d}{\gcd(g_i, d)} e_i^\top$ (with $g_i = 0$ read as $\gcd = d$, i.e. $\mu^\top = e_i^\top$), and $\lambda^\top = \mu^\top U$. Then $\lambda^\top A = \mu^\top UA = \mu^\top DV^{-1}$, and $\mu^\top D = \frac{d}{\gcd(g_i, d)} g_i e_i^\top = d \cdot \frac{g_i}{\gcd(g_i, d)} e_i^\top \equiv 0 \pmod{d}$, so $\lambda^\top A \equiv 0$. And $\lambda^\top s = \mu^\top s' = \frac{d}{\gcd(g_i, d)} s'_i$, which is $\not\equiv 0 \pmod{d}$ precisely because $\gcd(g_i, d) \nmid s'_i$. This is a certificate, giving (b). $\square$

Remark 12 (Why $d/\gcd(g_i, d)$ and not d/g_i). The invariant factor g_i need *not* divide d ; writing $u = (d/g_i)U_i$ would be ill-formed for composite d . The correct annihilator multiplier is $d/\gcd(g_i, d)$, which coincides with d/g_i only when $g_i \mid d$. [COMPLETE]

Lemma 13 (Integrality/gauge). *For even d and a closed commuting context C , the product $\prod_{v \in C} W(v)$ is an ω -power $\omega^{s(C)} I$ with $s(C) \in \mathbb{Z}_d$ canonically (no τ -half-steps). Kernel pairings $\lambda^\top s$ are independent of any admissible regauging of the $W(v)$.*

Proof. [COMPLETE] By Lemma 3 the product is $\tau^T I$ with $T = -\sum q(v_i) + 2\sum_{j < i} \beta(v_j, v_i)$. By the parity lemma (Lemma 26 of Part II) at even d every commuting pair has $c(v_j, v_i)$ even; telescoping, T is even, so $\tau^T = \omega^{T/2}$ and $s(C) = T/2 \pmod{d} \in \mathbb{Z}_d$. A regauging $W(v) \mapsto \zeta_v W(v)$ with $\zeta_v \in U(1)$ shifts $s(C) \mapsto s(C) + \sum_{v \in C} \theta_v$ ($\zeta_v = \omega^{\theta_v}$), i.e. $s \mapsto s + A\theta$; hence $\lambda^\top s \mapsto \lambda^\top s + (\lambda^\top A)\theta = \lambda^\top s$ for a certificate. $\square$

Theorem 14 (Detection equivalence). *Let d be even and $\mathcal{F}$ a closed commuting Weyl family with incidence matrix A (over $\mathbb{Z}_d$) and canonical phase vector $s \in \mathbb{Z}_d^k$ (Lemma 13). The following are equivalent:*

- (a) *no noncontextual $\mathbb{Z}_d$ -assignment exists (the system $A\gamma \equiv s \pmod{d}$ is infeasible);*
- (b) *some certificate λ ($\lambda^\top A \equiv 0$) has $\lambda^\top s \equiv d/2 \pmod{d}$;*
- (c) *some certificate has odd commutator carry, $\bar{\lambda} \cdot b \equiv 1 \pmod{2}$.*

Proof. [COMPLETE] (a) $\Leftrightarrow$ “ $\exists$ certificate with $\lambda^\top s \not\equiv 0$ ” is exactly Lemma 11 with the identifications $M = A$, phase = s . Given such a certificate, the Obstruction Spectrum Theorem 4 (which applies to *any* left-kernel vector, by the scope declared in §1 and Remark 7) forces $2(\lambda^\top s) \equiv 0$, hence the nonzero value is exactly $d/2$: this is (b). Trivially (b) $\Rightarrow$ (a) by mutual exclusion in Lemma 11. Finally (b) $\Leftrightarrow$ (c) is Corollary 6: $\lambda^\top s \equiv \frac{d}{2}(\bar{\lambda} \cdot b)$, which is $d/2$ iff $\bar{\lambda} \cdot b$ is odd. The canonical integral gauge needed to even state (a) is Lemma 13; kernel pairings are gauge invariant, so the equivalence is gauge free. $\square$

Remark 15 (What is new here vs. assembly). The three-way equivalence is five lines of ring theory (Lemma 11) bolted onto two proven theorems of Part I. The composite- d subtlety — that Fredholm over $\mathbb{Z}_d$ needs SNF and the $d/\gcd(g_i, d)$ multiplier — is genuine and fully discharged above. [COMPLETE]

5 The exact value-bit formula

We now promote the computational value-bit identity to a theorem. The setting is a *pure fiber pool*: a base family F at level d and, at level $2d$, observables $\tilde{v} = u + dx$ with $u = \tilde{v} \bmod d \in \{0, \dots, d-1\}^{2m}$ the base residue and $x = \tilde{v} \operatorname{div} d \in \{0, 1\}^{2m}$ the lift bits; a mod- $2d$ kernel element $\tilde{\lambda}$ with odd support supp .

Lemma 16 (Sum-of-floors carry identity). *For integers $M_1, \dots, M_n$ with $\Sigma = \sum_a M_a$ and $\kappa = \#\{a : M_a \text{ odd}\}$,*

$$\sum_a \left\lfloor \frac{M_a}{2} \right\rfloor \equiv \left\lfloor \frac{\Sigma}{2} \right\rfloor + \left\lfloor \frac{\kappa}{2} \right\rfloor \pmod{2}.$$

Proof. [COMPLETE] $\lfloor M/2 \rfloor = (M - (M \bmod 2))/2$ gives $\sum_a \lfloor M_a/2 \rfloor = (\Sigma - \kappa)/2$. Since $\Sigma \equiv \kappa \pmod{2}$, $\lfloor \Sigma/2 \rfloor = (\Sigma - (\kappa \bmod 2))/2$, and the difference is $((\kappa \bmod 2) - \kappa)/2 = -\lfloor \kappa/2 \rfloor \equiv \lfloor \kappa/2 \rfloor \pmod{2}$. $\square$

Lemma 17 (Exact per-pair lift carry). *With $q_u = \lfloor \langle u_a, u_b \rangle / d \rfloor$ replaced by $\langle u_a, u_b \rangle \bmod d \in [0, d)$ in the numerator, write $M_{ab} = \langle u_a, x_b \rangle + \langle x_a, u_b \rangle$, $s_{xx} = \langle x_a, x_b \rangle$. Then for a level- $2d$ commuting pair,*

$$\left\lfloor \frac{\langle \tilde{v}_a, \tilde{v}_b \rangle}{2d} \right\rfloor \equiv \left\lfloor \frac{(\langle u_a, u_b \rangle / d) + M_{ab} + d s_{xx}}{2} \right\rfloor \pmod{2},$$

a pure floor identity because $0 \leq \langle u_a, u_b \rangle \bmod d < d$ prevents the base remainder from carrying.

Proof. [SKETCH: CITED] Expand $\langle \tilde{v}_a, \tilde{v}_b \rangle = \langle u_a, u_b \rangle + dM_{ab} + d^2 s_{xx}$; divide by $2d$ and use that the pair commutes at level $2d$ (numerator $\equiv 0 \bmod 2d$) so the fraction is an integer, and that $\langle u_a, u_b \rangle = d\lfloor \langle u_a, u_b \rangle / d \rfloor + (\langle u_a, u_b \rangle \bmod d)$ with the remainder in $[0, d)$. The remainder cannot contribute a $2d$ -carry; the residual is the displayed floor. The boundary bookkeeping is verified exactly at $d = 4, 6, 8$ in `Wformula.py`. Cited: Paper B, App. A. $\square$

Theorem 18 (Value-bit formula). *For a mod- $2d$ kernel element $\tilde{\lambda}$ of a pure fiber pool with odd support supp , the value bit P (defined by $S = dP$) is*

$$P \equiv \frac{1}{2} \left[\sum_{\operatorname{supp}} K + \sum_{\operatorname{supp}} \Lambda(x) + \frac{d}{2} \sum_{\operatorname{supp}} Q(x) \right] \pmod{2},$$

where, per selected context, $K = \sum_{i < i'} \langle u_i, u_{i'} \rangle / d$, $\Lambda(x) = \langle \sum_i x_i, \sigma \rangle - \sum_i \langle x_i, u_i \rangle - 2 \sum_i \langle x_i, P_i \rangle$, $Q(x) = \sum_{i < i'} \langle x_i, x_{i'} \rangle$, $\sigma = \sum_i u_i$, P_i the base prefix sums. The bracket is always even, and the $(d/2)$ -terms survive mod 2 only for $d \equiv 2 \pmod{4}$.

Proof. [SKETCH: CITED] By Corollary 6 at level $2d$, $P = \bar{\lambda} \cdot b_{2d} = \sum_{C \in \text{supp}} b_{2d}(C) \pmod{2}$, and by Theorem 8 $b_{2d}(C) = \sum_{i < i'} \langle \tilde{v}_i, \tilde{v}_{i'} \rangle / (2d)$. Lemma 17 rewrites each summand as $\lfloor ((\langle u_i, u_{i'} \rangle / d) + M_{ii'} + d s_{ii'}) / 2 \rfloor$; Lemma 16 then converts the sum of per-pair floors into (i) $\lfloor (\sum_{i < i'} (\dots)) / 2 \rfloor$ and (ii) a parity-count carry $\lfloor \kappa / 2 \rfloor$. Grouping the base term into K , the cross term $\sum_{i < i'} M_{ii'}$ into Λ (the identity $\sum_{i < i'} M_{ii'} = \langle \sum x_i, \sigma \rangle - \sum \langle x_i, u_i \rangle - 2 \sum \langle x_i, P_i \rangle$ is bilinearity plus the prefix expansion), and the quadratic term $\sum s_{ii'}$ into $Q(x)$ with the explicit $d/2$ weight, yields the displayed bracket. Bracket evenness is equivalent to $P \in \mathbb{Z}$, which holds because $S \in \{0, d\}$ at level $2d$ (Theorem 4). The exact matching of the residual carry $\lfloor \kappa / 2 \rfloor$ against the stated coefficients is the content verified non-vacuously (0/550) in `Wformula.py`. *Cited:* Paper B, Thm. W. $\square$

Status of the step. Structure (reduction to $\sum b_{2d}$ and the three natural terms) is [COMPLETE]; the exact integer coefficients and bracket-evenness rest on Lemma 17 and are computationally certified, hence [SKETCH: CITED] overall. This is the one Part-I theorem not fully closed by hand here.

6 Attainment and the classification

Theorem 19 (Attainment / classification). *For every even d and every $m \geq 2$, the value $d/2$ is attained by a closed commuting Weyl family; combined with Theorem 4, $H(d, m) = \{0, d/2\}$ for even d and $H(d, m) = \{0\}$ for odd d .*

We prove attainment in three pieces of decreasing self-containment, and label each honestly.

(A) A complete parametric construction for $d \equiv 2 \pmod{4}$

Proposition 20 (Signed scaled Peres–Mermin). *Let $d \equiv 2 \pmod{4}$ and $\delta = d/2$ (odd). Let $\{v_1, \dots, v_9\} \subset \mathbb{Z}_d^4$ be the nine Peres–Mermin vectors on $m = 2$ qudits arranged in a 3×3 grid, with the six line-contexts (three rows, three columns). Scale every vector to $\delta v \in \mathbb{Z}_d^4$. Then the six scaled lines are closed commuting contexts, and the signed vector $\lambda = (+1, +1, +1, -1, -1, -1)$ (rows $+1$, columns -1) is a genuine $\mathbb{Z}_d$ -certificate with value $S \equiv d/2$.*

Proof. [COMPLETE] *Grid.* Take, in per-site coordinates (a_1, b_1, a_2, b_2) ,

$$\begin{aligned} X \otimes I &= (1, 0, 0, 0) & I \otimes X &= (0, 0, 1, 0) & X \otimes X &= (1, 0, 1, 0) \\ I \otimes Z &= (0, 0, 0, 1) & Z \otimes I &= (0, 1, 0, 0) & Z \otimes Z &= (0, 1, 0, 1) \\ X \otimes Z &= (1, 0, 0, 1) & Z \otimes X &= (0, 1, 1, 0) & Y \otimes Y &= (1, 1, 1, 1), \end{aligned}$$

rows and columns being the six contexts. Each of the six lines sums to an even vector $((2, 0, 2, 0), (0, 2, 0, 2), (2, 2, 2, 2), (2, 0, 0, 2), (0, 2, 2, 0), (2, 2, 2, 2))$, and within each line every pair has *even* integer symplectic product (the operators pairwise commute over $\mathbb{Z}_2$). Each of the nine observables lies in exactly one row and one column.

Closure and commutation after scaling. $\sum_{v \in \text{line}} \delta v = \delta \cdot (\text{even vector}) = d \cdot (\text{integer vector}) \equiv 0 \pmod{d}$; and for u, w in a common line $\langle \delta u, \delta w \rangle = \delta^2 \langle u, w \rangle = \frac{d^2}{4} \langle u, w \rangle$; since $\langle u, w \rangle$ is even, write $\langle u, w \rangle = 2w_{uw}$, so $\langle \delta u, \delta w \rangle = \frac{d^2}{2} w_{uw} = d \cdot (\delta w_{uw}) \equiv 0 \pmod{d}$. So the scaled lines are closed commuting contexts.

Certificate. Each observable is one row-edge and one column-edge of $K_{3,3}$, so $(\lambda^\top A)_v = \lambda_{\text{row}(v)} + \lambda_{\text{col}(v)} = (+1) + (-1) = 0$ exactly in $\mathbb{Z}_d$; λ is a genuine $\mathbb{Z}_d$ -cycle.

Value. By Corollary 6, $s(C) \equiv -\sum_{v \in C} q(v) + \frac{d}{2} b(C)$, so

$$S = \lambda^\top s = \sum_{\text{rows}} s - \sum_{\text{cols}} s \equiv -\left(\sum_{\text{rows}} \sum_{v \in C} q(v) - \sum_{\text{cols}} \sum_{v \in C} q(v) \right) + \frac{d}{2} \left(\sum_{\text{rows}} b - \sum_{\text{cols}} b \right) \pmod{d}.$$

Each observable appears once among rows and once among columns, so the two $\sum q$ totals are both $\sum_{\text{obs}} q(v)$ and cancel. For the carry, compute $b(\delta C)$: with $\langle \delta u, \delta w \rangle = \frac{d^2}{2} w_{uw}$ and $\langle \delta u, \delta w \rangle / d = \delta w_{uw}$,

$$b(\delta C) \equiv \sum_{u < w \in C} \frac{\langle \delta u, \delta w \rangle}{d} = \delta \sum_{u < w \in C} w_{uw} \equiv \sum_{u < w \in C} w_{uw} \pmod{2} \quad (\delta \text{ odd}),$$

and $w_{uw} = \langle u, w \rangle / 2$ is exactly the $d = 2$ carry summand, so $b(\delta C) \equiv b_{\text{PM}}(C) \pmod{2}$ context-by-context. Because the qubit Peres–Mermin square is contextual with value 1, the all-lines total carry is odd: $\sum_{\text{all } C} b_{\text{PM}}(C) \equiv 1$. Hence $\sum_{\text{rows}} b - \sum_{\text{cols}} b \equiv \sum_{\text{all}} b \equiv 1 \pmod{2}$, and $S \equiv \frac{d}{2} \cdot 1 = d/2 \pmod{d}$. $\square$

This is a fully self-contained attainment at *every* $d \equiv 2 \pmod{4}$ ($d = 2, 6, 10, 14, \dots$), with an explicit genuine $\mathbb{Z}_d$ -certificate.

(B) General even d : inherited from ROZF

Proposition 21 (General even- d attainment). *For every even d (in particular $d \equiv 0 \pmod{4}$), where construction (A) degenerates) there is a closed commuting Weyl family attaining $d/2$.*

Proof. [SKETCH: CITED] The elementary scaling of (A) fails for $4 \mid d$: then $\delta = d/2$ is even, every scaled carry $b(\delta C) \equiv \delta \sum w_{uw} \equiv 0 \pmod{2}$ vanishes, and the family is noncontextual — a genuinely different construction is required. Raussendorf–Okay–Zurel–Feldmann [1] construct, for every even d , an even- d cohomological (“face”) obstruction realizing the value $d/2$; their face construction is reported to attain $\{0, d/2\}$ at $d = 4, 6, 8, 10, 12, 16, 20$. We *inherit* the generality of attainment from [1] and only translate their data into the present certificate language (λ, A, s) , where the value $d/2$ then follows from Corollary 6. *Cited:* [1], Quantum **7**, 979 (2023). $\square$

Status of the step. Generality over all even d is [SKETCH: CITED], credited to [1]. What is *original* here: (i) the Weyl-certificate translation of the obstruction into (λ, A, s) with the value read off Corollary 6; (ii) the closed parametric family of (A) for $d \equiv 2 \pmod{4}$; and (iii) the explicit certificates of (C). What is *inherited*: the existence of an attaining family at $d \equiv 0 \pmod{4}$ in general.

(C) Explicit original certificates at $d = 8, 16$

Proposition 22 (Matrix-verified certificates). *There is an explicit closed commuting Weyl family at $d = 8$ (89 contexts) with a certificate of value $4 = d/2$, and one at $d = 16$ (363 contexts) with value $8 = d/2$. Both lie in the class $d \equiv 0 \pmod{4}$ that construction (A) cannot reach.*

Proof. [SKETCH: CITED] The certificates are exhibited and checked on raw 64×64 and 256×256 complex matrices: every listed context is verified pairwise commuting with scalar product, $\lambda^\top A \equiv 0 \pmod{d}$ holds, the value equals $d/2$, and the noncontextual system is infeasible (`verify_cert8.py`, `verify_cert16.py`). These certificates are new to this program and are independent of the tower results; the $d = 16$ instance shows there is no depth ceiling. The by-hand reduction of these finite objects to Corollary 6 is routine but the data are verified by machine, hence [SKETCH: CITED]. *Cited:* Paper B, §6. [COMPLETE] for the *logic* that a value- $d/2$ certificate implies attainment (Theorem 4, Corollary 6); [SKETCH: CITED] for the certificates’ numerical correctness. $\square$

7 Depth Rigidity

Definition 23 (Lift; “top 2-adic layer”). A *lift* of a level- d family to level $2d$ assigns $x_v \in \{0, 1\}^{2m}$ per observable, setting $\tilde{v} = v + dx_v$, subject to keeping every context commuting and closed at level $2d$. The mod-2 carry class $b \in \mathbb{F}_2^k / \text{rowspace}_{\mathbb{F}_2} A$ is the *top 2-adic layer* of the family: it is the obstruction visible at the current level and *not* inherited from any finer datum one level up. “No single lift carries the obstruction” means: the whole b -class is a coboundary of a per-observable function whenever any global lift exists, and conversely a genuine b -class forbids every global lift.

Theorem 24 (Depth Rigidity). *Fix even d and a closed commuting level- d family. Reducing the commutation constraints at level $2d$ gives, for each in-context pair (i, j) , the $\mathbb{F}_2$ -linear equation*

$$\langle v_i, x_j \rangle + \langle x_i, v_j \rangle \equiv k_{ij} := \frac{\langle v_i, v_j \rangle}{d} \pmod{2}, \quad (5)$$

and the closure equation $r_C + \sum_{v \in C} x_v \equiv 0 \pmod{2}$ with $r_C = \frac{1}{d} \sum_{v \in C} v$.

- (a) *If the commutation block (5) is solvable, then $b(C) \equiv \sum_{v \in C} \langle v, x_v \rangle \pmod{2}$ is a coboundary $b = Ah$ ($h_v = \langle v, x_v \rangle \pmod{2}$); hence every certificate of the lifted family has value 0.*
- (b) *If the level- d family attains ($\exists \lambda$ with value $d/2$), the $\bar{\lambda}$ -weighted commutation equations have identically zero left-hand side and right-hand side $\sum_r \bar{\lambda}_r b(C_r) \equiv 1$; the block is infeasible, so attaining families never lift.*

Proof. [COMPLETE] *Derivation of (5).* $\langle \tilde{v}_i, \tilde{v}_j \rangle = \langle v_i, v_j \rangle + d(\langle v_i, x_j \rangle + \langle x_i, v_j \rangle) + d^2 \langle x_i, x_j \rangle$. Commuting at level d gives $\langle v_i, v_j \rangle = dk_{ij}$; dividing $\langle \tilde{v}_i, \tilde{v}_j \rangle \equiv 0 \pmod{2d}$ by d and using $d^2 \langle x_i, x_j \rangle \equiv 0 \pmod{2}$ (even d) gives $k_{ij} + \langle v_i, x_j \rangle + \langle x_i, v_j \rangle \equiv 0 \pmod{2}$, i.e. (5). Closure is identical with $\sum \tilde{v} = d(r_C + \sum x_v) \equiv 0 \pmod{2d}$.

A per-context reduction. Work mod 2, where $\langle a, b \rangle \equiv \langle b, a \rangle$. For any context,

$$\sum_{i < j} (\langle v_i, x_j \rangle + \langle x_i, v_j \rangle) \equiv \sum_{i \neq j} \langle v_i, x_j \rangle = \langle \sum_i v_i, \sum_j x_j \rangle - \sum_i \langle v_i, x_i \rangle \pmod{2},$$

and $\sum_i v_i = dr_C$ makes $\langle \sum v_i, \sum x_j \rangle = d \langle r_C, \sum x_j \rangle \equiv 0 \pmod{2}$. Hence

$$\sum_{i < j \in C} (\langle v_i, x_j \rangle + \langle x_i, v_j \rangle) \equiv \sum_{v \in C} \langle v, x_v \rangle \pmod{2}. \quad (6)$$

(a). If (5) holds for a solution x , sum it over $i < j \in C$: the left side is (6) $= \sum_{v \in C} \langle v, x_v \rangle = (Ah)_C$ with $h_v = \langle v, x_v \rangle \pmod{2}$, while the right side is $\sum_{i < j \in C} k_{ij} = \sum_{i < j} \langle v_i, v_j \rangle / d = b(C)$ (Theorem 8). Thus $b = Ah$, a coboundary; any certificate $\bar{\lambda}$ ($\bar{\lambda}^\top A \equiv 0$) pairs to $\bar{\lambda} \cdot b = \bar{\lambda}^\top Ah = 0$, so all lifted values vanish.

(b). Take an attaining λ ; then $\bar{\lambda} \cdot b \equiv 1$ (Corollary 6). Weight (5) by $\bar{\lambda}_C$ and sum over all contexts and in-context pairs. The right side is $\sum_C \bar{\lambda}_C \sum_{i < j \in C} k_{ij} = \sum_C \bar{\lambda}_C b(C) = \bar{\lambda} \cdot b \equiv 1$. The left side, by (6), is $\sum_C \bar{\lambda}_C \sum_{v \in C} \langle v, x_v \rangle = \sum_v (\bar{\lambda}^\top A)_v \langle v, x_v \rangle \equiv 0$, because $\lambda^\top A \equiv 0 \pmod{d}$ reduces to $\bar{\lambda}^\top A \equiv 0 \pmod{2}$. So any x would give $0 \equiv 1$: the commutation block is infeasible and no lift exists. $\square$

Remark 25 (Reading of the rigidity). The obstruction bit is precisely the $\mathbb{F}_2$ -inconsistency of the global lift system, pinned by the *certificate* $\bar{\lambda}$ and not by any observable’s individual lift x_v (each x_v enters only through the coboundary $h_v = \langle v, x_v \rangle$, which pairs to 0 with every cycle). Whenever a global lift exists the obstruction trivializes; when it does not, the whole family is rigid. This is the exact sense of “top layer, carried by no section.” [COMPLETE]

Part II. The Local Valuation Theorem

8 Setup and the parity lemma

A single context has an isotropic (internally commuting) label group $G \subseteq \mathbb{Z}_d^{2m}$, $\langle u, v \rangle \equiv 0 \pmod{d}$ for all $u, v \in G$. The Weyl operators define a $U(1)$ -valued 2-cocycle on G ,

$$f(u, v) = \tau^{c(u, v)}, \quad W(u)W(v) = f(u, v) W((u + v) \bmod d),$$

with c as in Lemma 2. We say f *trivializes* (the context is *classical*) if there is a cochain $\mu : G \rightarrow U(1)$ with $f(u, v) = \mu(u)\mu(v)\mu(u + v)^{-1}$; then $V(v) := \mu(v)^{-1}W(v)$ is an honest (nonprojective) representation of G . The *valuation* of the context is the least N with an admissible μ valued in μ_N .

Lemma 26 (Parity lemma). *For all u, v , $c(u, v) \equiv \langle u, v \rangle + d \cdot \rho(u, v) \pmod{2}$ for an integer ρ ; hence for even d , $c(u, v) \equiv \langle u, v \rangle \pmod{2}$, and on a commuting pair at even d , $c(u, v)$ is even. For odd d the correction $d\rho$ may be odd, so $c(u, v)$ can be odd on a commuting pair.*

Proof. [COMPLETE] One qudit suffices (both sides add over sites). With $u = (a, b), v = (a', b')$ and carries $e_a, e_b \in \{0, 1\}$ defined by $a + a' = (a + a' \bmod d) + d e_a$, $b + b' = (b + b' \bmod d) + d e_b$, Lemma 2 gives $c(u, v) = -ab - a'b' + 2a'b + q((u + v) \bmod d)$. Expanding $q((u + v) \bmod d) = (a + a' - d e_a)(b + b' - d e_b)$,

$$c(u, v) = ab' + a'b - d(e_b(a + a') + e_a(b + b')) + d^2 e_a e_b.$$

Mod 2, $ab' + a'b \equiv ab' - a'b = \langle u, v \rangle$, so $c(u, v) \equiv \langle u, v \rangle + d\rho \pmod{2}$ with $\rho = e_b(a + a') + e_a(b + b') + d e_a e_b$. For even d the $d\rho$ term vanishes mod 2; on a commuting pair $d \mid \langle u, v \rangle$ with d even forces $\langle u, v \rangle$ even, so $c(u, v)$ is even. $\square$

Corollary 27 (Even- d contexts need no half- τ steps). *For even d , every closed commuting context has even τ -level phase: its product is an ω -power $\omega^{s(C)}I$, $s(C) \in \mathbb{Z}_d$, with integrality automatic (not a gauge choice). This is Lemma 13, now seen as an immediate consequence of Lemma 26: the telescoped exponent $T = \sum_i c(P_i, v_i)$ is a sum of even integers.* [COMPLETE]

9 The cyclic transgression

Lemma 28 (Cyclic power scalar). *Let $g \in G$ have order $a = \text{ord}(g)$ in $\mathbb{Z}_d^{2m}$ (so $a \mid d$). Then $W(g)^a = \tau^\Gamma I$ with*

$$\Gamma \equiv a(a - 2)q(g) \pmod{2d}.$$

Consequently any trivializing μ must satisfy $\mu(g)^a = \tau^\Gamma$, and one may take $\mu(g) = \tau^{(a-2)q(g)}\zeta$ with $\zeta^a = 1$; the essential factor is $\tau^{(a-2)q(g)} \in \mu_{2d}$.

Proof. [COMPLETE] Telescoping (Lemma 3 applied to the word $g, g, \dots, g$ of length a , whose sum is $ag \equiv 0$), $\Gamma = \sum_{k=1}^{a-1} c(kg \bmod d, g)$. Using $c(w, g) = -q(w) - q(g) + 2\beta(w, g) + q((w + g) \bmod d)$ with $w = kg \bmod d$: the terms $-q(kg) + q((k+1)g)$ telescope over $k = 1, \dots, a-1$ to $q(ag \bmod d) - q(g) = -q(g)$; the constant $-q(g)$ sums to $-(a-1)q(g)$; and $2 \sum_{k=1}^{a-1} \beta(kg \bmod d, g) \equiv 2q(g) \sum_{k=1}^{a-1} k = q(g)a(a-1) \pmod{2d}$ (as $\beta(g, g) = q(g)$ and reductions cost multiples of $2d$). Summing, $\Gamma \equiv -q(g) - (a-1)q(g) + a(a-1)q(g) = q(g)(-a + a(a-1)) = a(a-2)q(g) \pmod{2d}$. For a genuine rep $V(g) = \mu(g)^{-1}W(g)$ with $V(g)^a = V(ag) = I$ one needs $\mu(g)^a = W(g)^a = \tau^\Gamma$; the stated $\mu(g)$ solves this since $\mu(g)^a = \tau^{a(a-2)q(g)}\zeta^a = \tau^\Gamma$. $\square$

Lemma 29 (Divisibility / parity of the essential exponent). $(a - 2)q(g)$ is even for every g when d is even; for odd d it can be odd.

Proof. [COMPLETE] Since $a = \text{ord}(g) \mid d$ and $a q(g) = \sum_i (a g_{a,i}) g_{b,i} \equiv 0 \pmod{d}$ (each $a g_{a,i} \equiv 0$), we get $q(g) \equiv 0 \pmod{d/a}$. *Even d :* if a is even then $a - 2$ is even and $(a - 2)q(g)$ is even; if a is odd then d/a is even (as d is even, a odd), so $q(g)$ is even and again $(a - 2)q(g)$ is even. *Odd d :* a is odd, $a - 2$ is odd, and $q(g) \equiv 0 \pmod{d/a}$ with d/a odd allows $q(g)$ odd, so $(a - 2)q(g)$ can be odd. $\square$

10 The theorem

Theorem 30 (Local classicality, sharp valuation). *Every single Weyl context (commuting label group G) is classical: its cocycle f is a coboundary. A trivializing cochain μ can be chosen valued in*

$$\mu_d \quad (d \text{ even}), \quad \mu_{2d} \quad (d \text{ odd}),$$

and the odd- d bound is sharp: the group $\langle(1, 1)\rangle$ at $d = 3$ admits no μ_3 -valued trivializer.

Proof. [COMPLETE] *Cyclic factors.* Decompose $G = \bigoplus_\ell \langle g_\ell \rangle$ (structure theorem for finite abelian groups). On each factor take $\mu(g_\ell) = \tau^{(a_\ell - 2)q(g_\ell)} \zeta_\ell$ as in Lemma 28 and extend by $\mu(k g_\ell) = \mu(g_\ell)^k \tau^{-\Gamma_{\ell,k}}$, where $W(g_\ell)^k = \tau^{\Gamma_{\ell,k}} W(k g_\ell)$; this makes $V = \mu^{-1} W$ a representation of $\langle g_\ell \rangle$. By Lemma 29 the essential factor $\tau^{(a_\ell - 2)q(g_\ell)}$ lies in μ_d for even d ; and each $\Gamma_{\ell,k} = \sum_{j < k} c(j g_\ell, g_\ell)$ is a sum of *even* c -values (Lemma 26, commuting pairs, even d), so $\tau^{\Gamma_{\ell,k}} = \omega^{\Gamma_{\ell,k}/2} \in \mu_d$. Hence the whole cyclic cochain is μ_d -valued for even d , and μ_{2d} -valued for odd d .

Cross terms. The remaining data are the pairings f across distinct factors, cohomologous (via the global coboundary τ^{-q} , i.e. $W = \tau^{-q} V_0$ with $V_0 = X^a Z^b$ having cocycle ω^β) to the symmetric bilinear form $\beta|_G$ valued in $\mathbb{Z}_d$. A symmetric bilinear B on G has cocycle ω^B trivialized by any quadratic refinement $\psi : G \rightarrow \mathbb{Z}_d$ (resp. $\mathbb{Z}_{2d}$) with $B(u, v) = \psi(u) + \psi(v) - \psi(u + v)$; on $\bigoplus \langle g_\ell \rangle$ set $\psi(\sum_\ell k_\ell g_\ell) = \sum_\ell \psi_\ell(k_\ell) + \sum_{\ell < \ell'} k_\ell k_{\ell'} \beta(g_\ell, g_{\ell'})$, whose diagonal parts ψ_ℓ are the cyclic solutions above and whose cross parts are $\mathbb{Z}_d$ -valued (so μ_d -valued in either parity). Symmetric-cocycle classes are additive over the cyclic decomposition ($\text{Ext}(\bigoplus \mathbb{Z}_{a_\ell}, \mathbb{Z}_d) = \bigoplus \mathbb{Z}_{\text{gcd}(a_\ell, d)}$), so the assembled μ trivializes f globally with the stated valuation.

Sharpness at $d = 3$. Let $g = (1, 1)$ on one qudit, $d = 3$; then $a = \text{ord}(g) = 3$, $q(g) = 1$, and Lemma 28 gives $\Gamma \equiv 3 \cdot 1 \cdot 1 = 3 \pmod{6}$, so $W(g)^3 = \tau^3 I = e^{i\pi} I = -I$ (directly: $(XZ)^3 = \omega^3 I = I$ at $d = 3$ and $W(g)^3 = \tau^{-3} (XZ)^3 = -I$). Any trivializer obeys $\mu(g)^3 = W(g)^3 = -1$. But every element of μ_3 cubes to $1 \neq -1$; hence no μ_3 -valued μ exists, while $\mu(g) = \tau \in \mu_6 = \mu_{2d}$ works. The valuation is exactly μ_{2d} . $\square$

Status of the step. The cyclic factor, the parity lemma, the transgression/divisibility, and the $d = 3$ sharpness are each [COMPLETE]. The cross-term assembly invokes the standard additivity of symmetric extension classes over a cyclic decomposition (finite abelian group cohomology), used as a named structural fact but elementary; we mark the assembly [COMPLETE] with that reliance made explicit.

Corollary 31 (Genuine τ -steps are an odd- d phenomenon). *The half- τ ($\mu_{2d} \setminus \mu_d$) valuation occurs only for odd d , and there iff some cyclic factor has $(a - 2)q(g)$ odd (equivalently $\mu(g)^a = -1$). For even d no context ever requires a half- τ step. This inverts the naive expectation that “more phase” lives at even d . [COMPLETE]*

Remark 32 (Repeated generators / nonminimal presentations). The valuation is an invariant of the pair (G, f) , not of any generating set: μ is a function on the group G , and a redundant or nonminimal presentation only adds relations already satisfied by μ . Concretely, if a generator is repeated (or a relation $\sum n_\ell g_\ell = 0$ is imposed), the transgression consistency $\mu(0) = 1$ is automatic because Γ was computed from the intrinsic order $a = \text{ord}(g)$; adding g twice to a context word multiplies the phase by $f(w, g)f(w + g, g)$, an even τ -power at even d (Lemma 26), so no new half-step appears. Hence the μ_d/μ_{2d} dichotomy is stable under all presentations. [COMPLETE]

11 Status

Each row records whether this document contains a self-contained proof ([COMPLETE]), a sketch resting on a named external result ([SKETCH: CITED]), or an explicit open sub-lemma ([GAP]). “Abstract item” points to the corresponding Paper B abstract claim.

Result (this document)	Status	Remark
Weyl product law (Lem. 2), telescope (Lem. 3)	[COMPLETE]	the sole external input, derived here
Thm. 4 Obstruction Spectrum ($2S \equiv 0$; $\{0, d/2\} / \{0\}$)	[COMPLETE]	multiplicities, repeats, nonmaximal, disconnected all allowed (shown in proof)
Thm. 8 Commutator-carry form $b = \sum \langle u, v \rangle / d$	[COMPLETE]	<i>derived</i> from the cocycle, not asserted
Cor. 9 closed-form cycle criterion (Q)	[COMPLETE]	even- d ; inter-context terms vanish by closedness
Prop. 10 exact $\mathbb{Z}_d$ vs. $\mathbb{Z}_2$ shadow	[COMPLETE]/[GAP]	sharpness [COMPLETE]; Weyl-specific faithfulness of the shadow [GAP] (empirical)
Thm. 14 Detection equivalence (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c)	[COMPLETE]	incl. composite- d SNF double-annihilator (Lem. 11)
Thm. 18 Value-bit formula	[SKETCH: CITED]	structure [COMPLETE]; exact carry coefficients via Lem. 17 + <code>Wformula.py</code>
Thm. 19(A) attainment, $d \equiv 2 \pmod{4}$	[COMPLETE]	signed scaled Peres–Mermin, genuine $\mathbb{Z}_d$ -certificate
Thm. 19(B) attainment, general even d	[SKETCH: CITED]	generality inherited from [1]
Thm. 19(C) explicit $d = 8, 16$ certificates	[SKETCH: CITED]	original; numerically matrix-verified
Thm. 24 Depth Rigidity (a),(b)	[COMPLETE]	top-layer meaning made precise; no lift carries the bit
Thm. 30 Local valuation μ_d/μ_{2d} , sharp	[COMPLETE]	incl. analytic $d = 3$ sharpness ($\mu(g)^3 = -1$)
Cor. 27, Cor. 31, Rem. 32	[COMPLETE]	no half- τ at even d ; presentation-invariance

The most important remaining gap. The general even- d attainment for $d \equiv 0 \pmod{4}$ (Theorem 19(B)) is the only *existence* statement not proved here by an elementary parametric construction; it rests on [1]. The elementary construction (A) covers $d \equiv 2 \pmod{4}$ completely, and (C) gives explicit witnesses at $d = 8, 16$, but a uniform self-contained family for all $d \equiv 0 \pmod{4}$ is not established in this document. Secondly, Theorem 18’s exact carry-coefficient bookkeeping

(Lemma 17) is computationally rather than hand-certified, and Proposition 10’s claim that the $\mathbb{Z}_2$ carry shadow is *faithful* on Weyl families (not merely sound) is empirical. Everything else — the Spectrum Theorem, the carry form, detection equivalence with the composite- d argument, depth rigidity, and the sharp local valuation — is complete and self-contained.

References

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